



STRATA

Browse in context.

DOC 06 · BACKEND SCHEMA

Backend Schema

The local SQLite schema behind Continuum: profiles, windows, tabs, events, page states, Moments, bookmarks, downloads, and settings.

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strata — local-first browser
confidential — internal working document

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Storage Philosophy

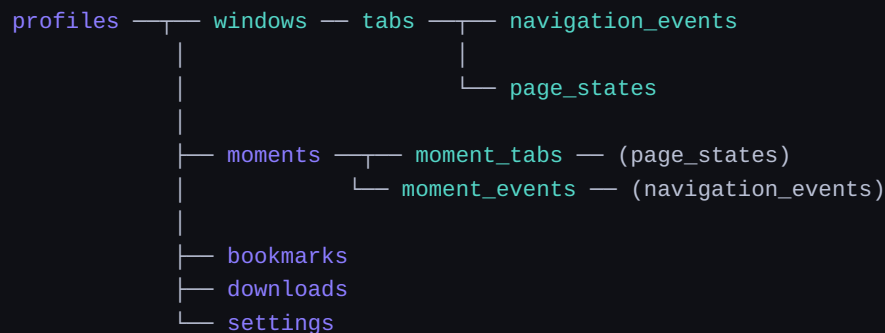
Strata uses **SQLite** as its single local datastore, keeping the product entirely local-first: no account, no server, no cloud, no analytics requirement. All data lives under one directory on the user's machine.

```
User's machine
  |
  ▼
~/ .strata/
  |
  ├── strata.db      ← all tables below
  ├── cache/        favicons, thumbnails
  ├── sessions/     transient/in-memory-adjacent session data
  └── moments/      optional per-Moment large assets (thumbnails)
```

PRIVACY DEFAULT

Continuum data never leaves `~/ .strata/` unless the user explicitly enables a future sync feature. Private browsing windows write nothing to `strata.db` at all — see §6.

Entity Overview



Every table below carries a `profile_id` back to **profiles** (directly, or indirectly through **tabs/windows**), so each of Personal / Work / Development / Guest keeps fully separate history, Moments, bookmarks, downloads, and settings, per the profile-isolation requirement in the TRD.

NOTE ON SCOPE

The product brief's suggested table list (§24) is followed exactly, with one pragmatic addition: **windows**. Tabs need a window to belong to in order to reconstruct multi-window layouts, and `WindowManager` already implies this table's existence — it is included here as a minimal, clearly-flagged extension rather than left implicit.

Browser State Tables

profilesIDENTITY & ISOLATION BOUNDARY		
COLUMN	TYPE	NOTES
id	TEXT PK	UUID
name	TEXT	"Personal", "Work", "Development", "Guest", or user-defined
kind	TEXT	personal · work · development · guest
is_ephemeral	BOOLEAN	true for Guest — cleared on quit, same rule as private mode
avatar_color	TEXT	hex, for the profile switcher UI
created_at	INTEGER	unix epoch seconds

windowsRECOMMENDED ADDITION — GROUPS TABS INTO A NATIVE WINDOW		
COLUMN	TYPE	NOTES
id	TEXT PK	UUID
profile_id	TEXT FK → profiles.id	
x	INTEGER	screen position
y	INTEGER	screen position
width	INTEGER	
height	INTEGER	
is_maximized	BOOLEAN	
is_fullscreen	BOOLEAN	
created_at	INTEGER	

tabs CURRENT, LIVE TAB STATE — NOT HISTORY		
COLUMN	TYPE	NOTES
id	TEXT PK	e.g. "tab_42"
profile_id	TEXT FK → profiles.id	
window_id	TEXT FK → windows.id	
url	TEXT	current URL
title	TEXT	
favicon_url	TEXT	cached under cache/
position	INTEGER	order within the tab bar
parent_tab_id	TEXT FK → tabs.id	nullable — tab relationship, e.g. opened-from
pinned	BOOLEAN	
muted	BOOLEAN	
is_active	BOOLEAN	the focused tab in its window
split_group_id	TEXT	nullable — tabs sharing this id are one split view
split_position	REAL	0.0–1.0, this pane's share of the split
created_at	INTEGER	
updated_at	INTEGER	

Continuum Tables

These four tables are Continuum's own storage: an append-only event log for Rewind, periodic state checkpoints, and the durable Moments library with its tab/event linkage.

navigation_events			EVENTRECORDER LOG — APPEND-ONLY, POWERS REWIND
COLUMN	TYPE	NOTES	
id	INTEGER PK	autoincrement	
tab_id	TEXT FK → tabs.id		
type	TEXT	TAB_CREATED · NAVIGATION · SCROLL · ZOOM_CHANGED · SPLIT_CREATED · MOMENT_SAVED · MOMENT_FROZEN · MOMENT_RESTORED	
url	TEXT	nullable — not all event types carry a URL	
title	TEXT	nullable	
metadata	JSON	type-specific payload, e.g. split ratio, zoom level	
timestamp	INTEGER	unix epoch seconds	

page_states

STATECOLLECTOR CHECKPOINTS — BROWSER-OWNED + BEST-EFFORT PAGE-OWNED STATE

COLUMN	TYPE	NOTES
id	INTEGER PK	autoincrement
tab_id	TEXT FK → tabs.id	
window_id	TEXT FK → windows.id	
url	TEXT	
title	TEXT	
favicon_url	TEXT	
scroll_x	REAL	browser-owned
scroll_y	REAL	browser-owned
zoom	REAL	browser-owned
navigation_index	INTEGER	position in this tab's back/forward stack
split_position	REAL	browser-owned, nullable
selected_text	TEXT	best-effort, nullable
focused_element	TEXT	best-effort, nullable
serialized_form_state	JSON	best-effort, nullable — never guaranteed
is_checkpoint	BOOLEAN	periodic snapshot vs. precise on-save capture
timestamp	INTEGER	

moments THE PERSISTENT, NAMED WORKSPACE SNAPSHOT		
COLUMN	TYPE	NOTES
id	TEXT PK	UUID
profile_id	TEXT FK → profiles.id	
name	TEXT	e.g. "React Authentication Research"
source	TEXT	explicit_save · freeze
window_layout	JSON	serialized window + split arrangement
thumbnail	TEXT	nullable path under moments/
tab_count	INTEGER	denormalized, powers homepage copy like "4 tabs"
created_at	INTEGER	
updated_at	INTEGER	bumped on rename or re-save

moment_tabs JOIN: TABS CAPTURED INSIDE A MOMENT, IN ORIGINAL ORDER/LAYOUT		
COLUMN	TYPE	NOTES
id	INTEGER PK	autoincrement
moment_id	TEXT FK → moments.id	
tab_order	INTEGER	restore order
url	TEXT	captured at save time
title	TEXT	
favicon_url	TEXT	
split_group_id	TEXT	nullable
split_position	REAL	nullable
page_state_id	INTEGER FK → page_states.id	the exact captured state to reapply on restore

moment_events			JOIN: NAVIGATION_EVENTS THAT ANCHOR/LED TO A MOMENT
COLUMN	TYPE	NOTES	
id	INTEGER PK	autoincrement	
moment_id	TEXT FK → moments.id		
navigation_event_id	INTEGER FK → navigation_events.id	cross-links Rewind timeline entries to the Moment they became	

Bookmarks, Downloads & Settings

bookmarks		
COLUMN	TYPE	NOTES
id	TEXT PK	UUID
profile_id	TEXT FK → profiles.id	
url	TEXT	
title	TEXT	
favicon_url	TEXT	
parent_id	TEXT FK → bookmarks.id	nullable, self-referential — folders
position	INTEGER	order within its folder
created_at	INTEGER	

downloads		
COLUMN	TYPE	NOTES
id	TEXT PK	UUID
profile_id	TEXT FK → profiles.id	
url	TEXT	source URL
file_path	TEXT	local destination
file_name	TEXT	
mime_type	TEXT	
size_bytes	INTEGER	
status	TEXT	in_progress · completed · failed · cancelled
started_at	INTEGER	
completed_at	INTEGER	nullable

settings			KEY/VALUE STORE, PER-PROFILE OR GLOBAL
COLUMN	TYPE	NOTES	
id	INTEGER PK	autoincrement	
profile_id	TEXT FK → profiles.id	nullable — NULL means an app-level global setting	
key	TEXT	e.g. "continuum.paused", "shortcuts.save_moment"	
value	TEXT	JSON-encoded value	
updated_at	INTEGER		

Write Patterns, Retention & Migrations

Async, batched, debounced writes

All Continuum writes are asynchronous and batched — the UI thread never blocks on a SQLite write. Scroll events specifically are debounced 500–1000ms client-side before a `page_states` checkpoint is even queued, per the snapshot strategy in the TRD (§4). An explicit Save Moment or Freeze bypasses the debounce window and writes immediately, since the user is actively waiting on that action to feel instant.

Retention

navigation_events

Append-only. Periodically compacted: entries older than a rolling window are summarized into `page_states` checkpoints and the raw event rows are pruned, keeping the event log from growing unbounded while Moments (which snapshot the data they need) remain unaffected.

page_states

Non-checkpoint rows are pruned on the same rolling schedule; rows referenced by a `moment_tabs.page_state_id` are retained indefinitely, since deleting them would break a saved Moment.

Private mode & deletion

- A private window's tabs, navigation, and state never reach `navigation_events` or `page_states` — writes are simply skipped for the duration of that window.
- "Clear history" deletes all `navigation_events` and non-Moment-referenced `page_states` for the active profile; Moments are untouched unless the user separately deletes them.
- Deleting a Moment cascades to its `moment_tabs` and `moment_events` rows; any `page_states` rows left with no remaining reference become eligible for the normal retention sweep.

Migrations

Schema changes ship through `src-tauri/src/storage/migrations.rs` as ordered, versioned migration files, applied on startup against a small internal schema-version marker. Every migration in this document set's SDLC review gate (see SDLC §3) must include a forward migration and a rollback note before merge.